Lesson 2: Data Visualization and Operational Dashboards

A clean Key Performance Indicator (KPI) table is an answer to one question at a time. A dashboard is an answer to "how is the operation doing, right now, at a glance?" The difference is not cosmetic. A reader with three minutes before a standup should be able to open a well-built operational dashboard and come away knowing whether anything is on fire, what the trend is, and what needs attention this week. A KPI table requires reading and interpretation. A dashboard does the interpretation visually and leaves the reader with attention to spend on decisions instead of arithmetic.

You are still working with the OrbitCo support team's 30-day data — the `Tickets` and `Uptime` Tables you cleaned in Lesson 1, and the KPI summary you built from them. In this lesson you turn that foundation into a one-page operational dashboard: a KPI tile strip at the top, a trend chart of incidents per week, a severity-mix breakdown, a conditional-formatted hotspot table, and traffic-light status indicators tied to Service Level Agreement (SLA) thresholds. This dashboard is the artifact you will embed in the briefing deck in Lesson 4, and it is the closest thing in this course to the On-the-Job Learning (OJL) 5d competency — **develop and maintain technical documentation, Standard Operating Procedures (SOPs), and runbooks** — because a live dashboard is living operational documentation.

Data visualization is a standard Registered Training Institute (RTI) topic for this course, and this lesson is where it lives explicitly. The goal is not to make the dashboard look impressive. The goal is to build visuals that answer the right questions, in the right form, without distorting the underlying truth. The most common failure in operational reporting is not a lack of charts; it is charts that look confident while quietly misleading the reader. Half of this lesson is about picking visuals that fit the question. The other half is about noticing when a visual is lying.

Learning Objectives

By the end of this lesson, you will be able to:

1. Select the appropriate chart type for a given data-communication goal (comparison, trend, composition, distribution)
2. Create pivot tables and pivot charts to summarize operational data
3. Build a one-page operational dashboard combining KPI summaries, trends, and status indicators
4. Use conditional formatting as a lightweight visualization technique (heat maps, data bars, icon sets)
5. Export Excel charts and dashboards for use in documentation and presentations
6. Identify and fix misleading visualizations (truncated axes, chart-type mismatch, 3D distortion)

Chart Types and When to Use Them

Every chart answers a specific kind of question. Choosing the chart type is choosing what question the reader is going to be able to answer from it. If you pick the wrong type, the chart is not wrong — it is irrelevant. The five chart types below cover almost every operational reporting case you will meet on the job.

Column and Bar Charts: Comparing Categories

A **column chart** (vertical bars) or **bar chart** (horizontal bars) compares a measured value across a small number of discrete categories. The axis of categories has no inherent order — severity, team, region — so any order is legal, though you should sort by value for readability. Use a bar (horizontal) chart when category labels are long; otherwise a column chart reads faster because the eye scans left-to-right across bases that sit on the same baseline.

For OrbitCo: a column chart of "ticket count by severity" (High, Medium, Low) is the right tool because the three categories are discrete and the question is a direct comparison. A column chart of "ticket count by day-of-week" also works because the seven categories are finite and unordered in any meaningful sense.

Line Charts: Trends Over Time

A **line chart** shows a continuous metric over ordered time. The x-axis must be ordered (day, week, month). The line implies connection between adjacent points, so only use it when that implication is honest — daily ticket counts, yes; annual ticket counts across unrelated companies, no.

For OrbitCo: a line chart of "daily new tickets" across the 30-day window is the canonical view. So is "weekly MTTR" across twelve weeks if the data window were larger. The line lets the reader see whether the metric is trending up, trending down, or wobbling around a stable mean.

Pie and Donut Charts: Composition of a Whole

A **pie chart** shows how a total breaks down into parts. The parts must add to 100% and there should be at most five slices. A pie with twelve slices is noise; a bar chart of the same twelve categories is readable.

For OrbitCo: a small donut chart showing the severity mix (High/Medium/Low as percentages of total) is appropriate because there are three slices and the question is explicitly compositional. A pie of "tickets by resolution code" with eight slices is not — switch it to a horizontal bar chart sorted by count so the reader can compare magnitudes directly.

Scatter Plots: Correlation Between Two Continuous Variables

A **scatter plot** puts two continuous numeric variables on the two axes and plots each record as a dot. The question it answers is "do these two things move together?"

For OrbitCo: a scatter plot with `TicketCountThatDay` on the x-axis and `AverageMTTRThatDay` on the y-axis asks whether heavy-volume days correspond to slower resolution — a real operational hypothesis worth checking before staffing decisions. If the dots rise together, volume is hurting quality; if they are uncorrelated, the team is scaling cleanly.

Combo Charts: Two Related Metrics on One View

A **combo chart** draws two series on the same chart with different chart types, often one as columns and one as a line, on two y-axes. Use it when two related metrics share a time axis but have different units.

For OrbitCo: a combo chart of "tickets per week" (columns, left axis, raw counts) and "MTTR per week" (line, right axis, minutes) is a one-visual answer to "is volume going up, and is response quality holding?" The left axis is counts, the right axis is minutes, and the reader gets both stories without flipping between two charts.

Rule of Thumb: Match Chart to Question

The shortest way to pick a chart is to name the question the visual must answer:

- Comparison across categories → column or bar
- Trend over time → line
- Composition of a whole → pie (up to 5 slices) or stacked bar
- Correlation → scatter
- Two metrics, shared time axis, different units → combo

If you cannot name the question, the chart is decoration. Decoration is the root cause of half the misleading visualizations in business reporting — people add charts because the deck "needs" one, not because the chart answers a real question.

Pivot Tables and Pivot Charts

Pivot tables are the tool that separates the person who can answer one question from the person who can answer twenty questions from the same dataset in five minutes. Every pivot is a reshaping of the underlying data — you tell Excel what to put on the rows, what to put on the columns, and what to aggregate in the middle — and the result is a summary that you could have produced with formulas but would not want to maintain by hand.

When to Use a Pivot vs Formulas

Formulas (`COUNTIF`, `AVERAGEIF`, `SUMIFS`) are right when you know exactly which KPIs you want and they will not change. They produce a fixed, formatted KPI table suitable for embedding in a report that someone else will read. The output is stable and the layout is yours.

Pivot tables are right when you are exploring — "what do the tickets look like by team and severity? what about by team and month? what if we subtotal by resolution code?" — or when the grouping dimensions are going to change as the audience asks follow-ups. A production dashboard typically uses both: formulas for the top-line KPI tiles, pivots behind the drill-down views where the reader may want to slice the data differently.

Anatomy of a PivotTable

Select any cell inside the `Tickets` Table and choose **Insert → PivotTable**. A new sheet opens with an empty pivot and the PivotTable Fields pane on the right. The four drop zones are:

- **Rows** — fields that become the row headers (e.g., Severity)
- **Columns** — fields that become the column headers (e.g., Month)
- **Values** — the aggregated numeric field (e.g., TicketID counted, or ResolutionMinutes averaged)
- **Filters** — fields that restrict what the whole pivot includes

Dragging `Severity` to Rows, `Team` to Columns, and `TicketID` to Values (aggregated as Count) gives you a crosstab of ticket counts by severity and team. Swap fields and the shape changes instantly; the underlying data does not move. That immediacy is the argument for pivots over formulas when you are still figuring out which question matters.

Value Field Settings

Click the drop-down next to a field in Values and choose **Value Field Settings** to change the aggregation. Count, Sum, Average, Max, and Min are the common five, and the very useful **Show Values As → % of Grand Total** (converts counts to percentages across the whole pivot) or **% of Row Total** (percentages within each row) are the two "display" transforms that operations analysts reach for most often.

For OrbitCo: a pivot with Severity on rows and `TicketID` on values, with the aggregation set to Count and the display set to "% of Grand Total," is the data behind the severity-mix donut chart on the dashboard.

Pivot Charts

Insert → PivotChart creates a chart bound to the pivot. The chart follows the pivot as you pivot — pivot by month and the chart redraws by month. This is the main reason to use a pivot chart over a regular chart: interactivity comes for free, and there is no re-wiring when the analysis question shifts.

Slicers and Timelines

A **slicer** (Insert → Slicer) is a button-panel filter for a pivot. Drop a slicer on `Team` and clicking `Platform` filters the pivot to Platform tickets only. A **timeline** is the same idea for a date field with a nice month/quarter/year interface. Slicers and timelines let a non-analyst explore the dashboard without learning the PivotTable Fields pane, which is the single biggest usability win for dashboards shared outside the analytics team.

Refreshing

Pivots do not update automatically when the source Table grows. **PivotTable Analyze → Refresh** (or right-click → Refresh) is how you pull in new rows. The dashboard's "refresh" button in Lesson 4 is effectively this click, and a common operational habit

is to pin a Refresh All shortcut on the Quick Access Toolbar so the full workbook updates in one action before a weekly review.

Activity: Build the OrbitCo Operational Dashboard

In this hands-on lab, you start from the cleaned workbook produced in Lesson 1 (`Tickets` Table, `Uptime` Table, `SLALookup` Table, `outputs` sheet) and build a one-page operational dashboard that combines KPI tiles, a trend chart, a composition chart, and a conditionally-formatted details band. This dashboard is the artifact you will embed in the capstone briefing in Lesson 4, and it is the single best demonstration piece of the OJL 5d competency in the course.

Building an Operational Dashboard

A **dashboard** is a single sheet that consolidates the most-read operational views into one layout. The reader should not need to scroll, switch tabs, or run a filter to get the main story. Everything they need is on one screen, and everything else is one click away.

The Standard Layout: Tiles → Trends → Details

The three-band dashboard is the most reliable layout. It works for ops dashboards, financial dashboards, and executive briefings with equally good results because it matches the way readers actually scan a page — headlines first, trends second, evidence third.

- **Top band (KPI tiles):** four to six single-value summaries — the headline numbers. For OrbitCo: total tickets, MTTR, SLA breach count, uptime %, open critical tickets.
- **Middle band (trends):** the charts that show how the KPIs evolved — a line chart of daily tickets, a combo chart of weekly volume vs MTTR, a donut of severity mix.
- **Bottom band (drill-down details):** a filterable table with the row-level data behind the KPIs — the `Tickets` Table itself, conditionally formatted so breaches jump out.

KPI Tiles

A KPI tile is a cell (or a shape containing a cell reference) that displays a single headline number with a label. The cleanest pattern is one cell for the label ("MTTR"), one cell below it for the value, formatted large, using a formula like `=AVERAGE(Tickets[ResolutionMinutes])`. For a more styled version, use a shape with text linked to a cell, or a single cell containing `=TEXT(AVERAGE(Tickets[ResolutionMinutes]),"0")&" min MTTR"`.

The `&` concatenation operator joins text and values — useful for self-labeling tiles. `TEXT(value, format)` applies a format string so you can control decimal places, units, and percentages without changing the underlying value.

Trend Lines and Sparklines

A **sparkline** is a mini-chart that lives inside a single cell. For KPI tiles, sparklines are the fastest way to add context — the MTTR tile can carry a sparkline of the last 30 days of daily MTTR right next to the headline number, so the reader sees both the current value and the trend without a dedicated chart. **Insert → Sparkline → Line** on a range of daily values places the sparkline in the target cell.

Status Indicators and Traffic Lights

A **traffic-light indicator** is a conditional-formatting rule that colors a tile green, yellow, or red based on thresholds. For the OrbitCo uptime tile, the rule is:

- ≥ 99.9% → green
- 99.5% to 99.9% → yellow
- < 99.5% → red

Use **Conditional Formatting → Icon Sets → 3 Traffic Lights** and set the thresholds in the rule editor. The result is a tile that communicates "we are meeting SLA" or "we are not" before the reader even reads the number.

Dashboard Hygiene

A few rules keep dashboards readable over months of use:

- **No raw data on the dashboard sheet.** The `Tickets` Table and `Uptime` Table live on their own sheets. The dashboard only pulls computed values.
- **Consistent formatting.** Same font, same color palette, same number formats across all tiles. Cosmetic variance reads as noise and makes the reader work to separate signal from presentation.
- **Named ranges over cell addresses.** If a formula on the dashboard reads `=Tickets[@Column]` or `=KPIs!$B$7`, it keeps working when the source grows. If it reads `=Sheet2!D428`, it silently breaks the next time someone inserts a row.
- **A timestamp.** Every dashboard should have a cell showing "Data as of: [date]" so the reader knows how fresh the view is. A dashboard without a freshness marker is a dashboard nobody trusts after a month.
- **A short `README` tab.** Describe what the dashboard is for, what the source data is, how to refresh it, and who to ask about it. Dashboards outlive their authors; the

`README` is how the next person maintains it.

The last three rules are the `OJL 5d` point exactly: a dashboard built this way is documentation, not a disposable view. The next operations analyst who inherits this workbook in two years should be able to keep it running without calling you.

Conditional Formatting as Visualization

Conditional formatting in Lesson 1 was about flagging anomalies. In Lesson 2 it is about visualization — using color and in-cell graphics to turn dense tables into something readable at a glance. The three rule families below each earn a place on the OrbitCo dashboard.

Heat Maps (Color Scales)

A **color scale** applies a gradient across a range of values — typically red (high) through yellow (middle) to green (low), or the reverse. Heat maps are right when you have a two-dimensional table of numbers (e.g., ticket count by team across twelve weeks) and the reader needs to spot the hot cells without reading all 96 numbers.

For OrbitCo: a heat map of weekly ticket counts by team (rows = teams, columns = weeks) instantly shows which team-week combinations had a surge. The hotspots jump off the page; the quiet cells recede into the background where they belong.

Data Bars

A **data bar** is an in-cell horizontal bar whose length is proportional to the value. Data bars are right when you have a single column of numbers and the reader needs to compare magnitude across rows without reading each number.

For OrbitCo: data bars on the `ResolutionMinutes` column of a severity-filtered view give an immediate sense of which tickets took the longest, without requiring a sort or a chart. The bar inside the cell is doing the visual work; the number is still there for precise lookup.

Icon Sets

Icon sets (up/down arrows, traffic lights, check/x) place a small icon next to the value based on a threshold rule. Icons are loud — a few in the right places are useful; a dashboard full of them becomes noise.

Be cautious with color-only icon sets (red/green especially) for accessibility — roughly 8% of men and 0.5% of women are red-green colorblind. Use shape differences (up-arrow vs down-arrow) along with color, not color alone, so the meaning survives a monochrome printout or a colorblind reader.

When Conditional Formatting Beats a Chart

A chart adds structure and an axis; conditional formatting adds visual cues to data the reader is already looking at. When the table is small, when space is tight, when the reader needs row-level detail alongside the visual summary, conditional formatting is faster to build, faster to read, and faster to maintain. The OrbitCo dashboard uses both — charts in the middle band, conditional formatting on the details band — not because one is better, but because each fits its band.

Avoiding Misuse

The most common mistake is stacking too many rules on one sheet. A dashboard with eight conditional-formatting rules applied across the same region becomes visual noise — every cell is colored, so no cell stands out. Aim for two to four rules per dashboard, each with a specific operational meaning. Anything more and you are diluting the signal you worked to surface.

Exporting Charts for Use Elsewhere

A dashboard lives in Excel, but the people it informs often do not open Excel. Exporting charts correctly into slides, docs, and wikis is the last mile of operational reporting, and the paste mode you choose decides whether the chart stays honest in the target document.

Copy as Picture vs Copy as Embedded Object

Copying a chart from Excel and pasting into PowerPoint offers three behaviors, and the right one depends on who else will edit the target file.

- **Paste as Picture** — a static image. Preserves the exact rendering, cannot be edited or updated. Right for final decks that will be printed or distributed.
- **Paste as Embedded Object** — pastes an editable Excel object inside the PowerPoint. The receiver can edit the chart but the source link is broken. Bloats file size and risks format drift as the chart styles in the deck diverge from the source workbook.
- **Paste with Link** — the chart stays connected to the source Excel file and updates when the source updates. Right for recurring status slides (e.g., a weekly operations review deck).

Linked Paste: When It Helps, When It Breaks

Linked paste is powerful and fragile. It helps when the Excel file lives in a stable path, the deck is updated on a schedule, and the same person owns both files. It breaks when the Excel file is renamed or moved, the deck is shared with someone who does not have access to the source, or the Excel file is edited on a different machine than the one where the link was created.

A practical rule: if you will not be the one opening the deck when it is used, paste as picture.

Saving a Chart as an Image

For documentation wikis, tickets, and emails, right-click the chart and choose **Save as Picture** (or select, copy, and paste into an image editor). Portable Network Graphics (PNG) preserves sharpness on transparent backgrounds; Joint Photographic Experts

Group (JPEG) is smaller but lossy. Always use PNG for charts with text.

Practical Rule

If it needs to be editable by others, paste linked. If it needs to be static and portable, paste as picture or save as PNG. That single heuristic covers almost every situation you will meet in day-to-day operations reporting.

Activity: Chart Critique

In this short group exercise, you review a pack of five pre-built charts drawn from operational data similar to OrbitCo's. Several of the charts contain design problems — truncated y-axes, 3D distortion, chart-type mismatches, misleading color — and one is genuinely well-designed. You will identify the problem in each case and propose a specific fix. The goal is not to be harsh; the goal is to build the critical-eye habit you will apply to every chart you ship from here forward, including the ones in your own dashboards.

Avoiding Misleading Visualizations

A chart that misleads is worse than no chart, because it creates false confidence. The patterns below show up often enough in business reporting that every operations analyst needs to recognize them — both to avoid them in their own work and to push back when someone else's chart is quietly lying.

Truncated Y-Axis

Starting a bar or column chart at a value other than zero exaggerates differences. A column chart of "uptime percentage" that starts at 99% makes a drop from 99.9% to 99.7% look catastrophic, when the real difference is 0.2%. Column and bar charts must start at zero unless the truncation is clearly marked and justified. Line charts have more flexibility because the reader interprets slopes, not bar heights, but truncation still affects perception — use it deliberately or not at all.

3D Charts

3D bar, column, and pie charts distort the data by introducing perspective. A slice in the front of a 3D pie looks larger than an equal slice in the back. A 3D column that tilts toward the viewer looks taller than one that tilts away. Modern reporting style drops 3D entirely — it adds no information and introduces distortion.

Chart-Type Mismatch

Using a line chart for unordered categories implies connection where none exists. Using a pie chart for a dozen categories produces unreadable slivers. Using a column chart for a time series loses the connectedness that makes the trend readable. These are the cases where the discipline of "name the question" from the first section pays off — a mismatched chart fails the naming test immediately.

Misleading Color

Using red and green for categorical (non-ordinal) data exploits the viewer's instinct that red = bad and green = good when no such meaning exists. It also hurts colorblind readers. Use a neutral palette for categorical data and reserve red and green for status (pass/fail, breach/ok).

Y-Axis Zero Rule

For bar and column charts, start the y-axis at zero. For line charts, start at zero when the metric is absolute (ticket counts), but allow non-zero starts for rates (percentages, latencies) where the useful range is narrow — with the caveat that you should clearly label the axis. A non-zero baseline is never wrong in itself; it is only wrong when it is hidden.

Wrap-Up

A dashboard earns its place by turning a clean dataset into a readable operational view — KPI tiles for the headlines, trend charts for the direction, status indicators for whether we are inside or outside of target, and a details band for the evidence. The OrbitCo dashboard you build in this lesson is the visual half of the capstone. Lesson 3 moves the work into PowerPoint; Lesson 4 turns the dashboard into a five-minute briefing that leadership can read and act on.

Key Takeaways

- Every chart answers a specific question — comparison, trend, composition, correlation, or two-metric combo. If you cannot name the question, the chart is decoration and probably misleading.
- Pivot tables are the fastest way to answer twenty variations of the same question; slicers and timelines make those variations available to non-analysts without a tour of the PivotTable Fields pane.
- An operational dashboard uses the three-band layout (tiles → trends → details), hides raw data on other sheets, carries a freshness timestamp, and ships with a short `README` tab so it outlives its author.
- Conditional formatting — color scales, data bars, icon sets — is a visualization tool in its own right. Two to four rules per dashboard, each with a specific operational meaning, beats eight competing rules every time.
- Misleading visualizations (truncated axes, 3D distortion, chart-type mismatch, misleading color) create false confidence. Recognizing them is as important as avoiding them, because someone else's chart will eventually show up in your inbox.

Reflection Questions

1. Think of the last dashboard you opened at work or school. What was the one thing it was trying to tell you, and did the layout make that thing obvious? What would you change about the tiles, the trends, or the details band?
2. Pick one chart you produced in the last month. Could a reader pass the "name the question" test on it? If not, which of the five chart types from this lesson fits the

real question better?

3. You inherit an Excel dashboard with no `README` tab, no timestamp, and a rat's nest of cell-address formulas. What is the first thing you change, and why?